Which process faults can be told apart? Diagnosability limits of in-situ monitoring in laser powder bed fusion

Amit Jain[a,*]

[a][Affiliation to be completed], [City], [Country]

Abstract

In-situ monitoring is standard on metal additive manufacturing systems, and classifiers report high accuracies for detecting defects from photodiode and imaging signals. Those results answer a question about outcomes: which defect formed. Specifying hardware requires an answer one step upstream — which process *causes* a sensor configuration can separate in principle, and which remain confounded whatever classifier is applied. Where two causes produce signatures differing by less than the sensor noise, no volume of training data distinguishes them; the limit is structural, not algorithmic. This paper applies structural and quantitative diagnosability analysis, mature in automotive and aerospace process control, to laser powder bed fusion. A calibrated reduced-order thermal model generates the response of a coaxial suite to five injected faults: laser power and scan speed deviation, thin powder layers from recoater short-feed, degraded shielding gas flow, and geometry-driven heat accumulation, with separability measured as an angle between noise-whitened fault signatures. Laser power and gas degradation prove *exactly* confounded instantaneously, by algebra rather than noise, so no instrument reading the melt pool separates them, and seven candidate observables raise the attainable rank only from three to four. Regressing the same three sensors onto known build context raises the rank to five and the worst pair from 0.00° to 25.83° at no hardware cost, outperforming two idealised direct measurements of the causes themselves. Diagnosability matrices are reported for five configuration classes typical of commercial machines.

Keywords: Additive manufacturing, Process monitoring, Diagnosability, Fault diagnosis, Laser powder bed fusion

1. Introduction

Laser powder bed fusion (LPBF) builds a part by melting thousands of short track segments per layer, and a modest excursion in any of several independent inputs can produce a rejected part. Machine manufacturers have responded by fitting in-situ monitoring as standard — coaxial photodiodes, melt pool cameras, pyrometers, and increasingly acoustic and powder-bed imaging — and a large literature maps those signals onto defects with high reported accuracy.

Almost all of that work answers a question about *outcomes*: given the signals, which defect formed? That matters for part acceptance, but it is not the question a machine builder asks when specifying hardware, nor the one a process engineer asks when a build drifts. Those concern *causes* — the laser is underdelivering, the recoater is short-feeding, the gas flow has degraded — and which it is determines what gets corrected.

The distinction is not semantic, because causes can be mutually indistinguishable in a way outcomes are not. If two causes drive the readings along the same direction in signal space, the readings contain no information separating them, and no classifier, however trained, can recover the difference. The limitation is structural — a property of the measurement and the physics, not of the algorithm. Conversely, a pair separable in principle but misclassified in practice is an algorithm that can be improved.

Diagnosability analysis is the established discipline for deciding which case holds. Developed within model-based fault detection and isolation (FDI), it asks, before any estimator is built, which faults a given measurement set can detect and which it can isolate from one another [1, 2]. The methods are standard where sensor sets are costly and must be justified in advance, but to the authors' knowledge they have not been applied to LPBF.

This paper applies them. A calibrated reduced-order thermal model is driven by five injected fault modes, the resulting signatures are compared pairwise against noise, and the outcome is a diagnosability matrix per configuration, dividing fault pairs into those resolvable from existing signals, those needing an added measurement, and those structurally irreducible. The contributions are: what the authors believe to be the first diagnosability matrix for the process; a demonstration that laser power and shielding gas degradation are *exactly* confounded instantaneously, so no melt-pool instrument separates them; the finding that the binding limit is the melt pool rather than the instrumentation; and the demonstration that build structure already present dissolves that limit at no hardware cost, outperforming two idealised direct measurements of the causes.

*Corresponding author.
Email address: amitkjain23@gmail.com (Amit Jain)

2. Background and related work

2.1. What in-situ monitoring classifies

Melt pool monitoring with coaxial photodiodes and cameras is the most widely deployed configuration, and its canonical application is flaw classification. Scime and Beuth identified signatures indicative of keyholing porosity and balling from high-speed visible imaging [3], and later work extended this across modalities and alloys, including mechanical-property prediction from coaxial signals [4]. In each case the predicted quantity is an outcome or a material state, and whether two distinct excursions could have produced the same signal is not posed.

A smaller body of work estimates process inputs, and is closer to the present question. RAISE-LPBF reconstructs laser power and scan speed independently from on-axis video on 316L, sampling both per scan line from continuous distributions so interactions can be studied [5]. Reijonen reports coaxial photodiode response to power, speed, focal position, scan orientation and shielding gas flow, finding power approximately linear, speed a power law, and the remainder non-linear and interactive [6]. These characterise *sensitivity*; identifiability — whether sensitivities to different causes are linearly distinguishable — is the question neither addresses.

Sensing that observes the powder bed forms a third class, attacking causes the melt pool reports only indirectly: Liu et al. recover local powder layer thickness and thermal transport from the post-recoat heat-up transient [8], and Bugatti and Colosimo detect geometric and surface anomalies below 100 µm with a recoater-mounted contact image sensor [9]. Both need added hardware and neither is simulated here; they appear in Section 4.7 as targets, not modelled options.

2.2. Lumped process metrics are incomplete

A recurring pattern is to report a *lumped* process quantity in place of its components. Volumetric energy density (VED) is the usual choice, and state-of-the-art parameter estimation from melt pool imagery still predicts energy density *class* rather than power and speed separately [10].

There is direct evidence that this discards information. Guo et al. held input energy density constant while varying power and speed together, and observed melt pool dimensions and volume changing by orders of magnitude with transitions between melting regimes, attributing it to the two altering absorption differently through the vapour depression [11]. Independently, the 2022 Additive Manufacturing Benchmark found single-track depth correlated with VED while width did not [12]. Falling back on a lumped group is the natural response to confounded components; whether they are confounded for a given suite is a diagnosability question, and is what the literature has not established.

2.3. Diagnosability analysis

Model-based diagnosability analysis operates on a model expressed over known variables, unknown variables and faults. Structural analysis considers only which variables appear in which equations, discarding numerical values, and determines detectability and isolability from the resulting bipartite graph; minimal structurally overdetermined (MSO) sets identify the redundant subsystems from which consistency tests are built [1, 2]. Being purely combinatorial it scales to large nonlinear models and is insensitive to parameter error, but it can be optimistic: two faults may be structurally isolable yet numerically indistinguishable. Quantitative analysis supplies the remainder, by linearising the fault-to-measurement map and examining the geometry of the resulting signature directions. This paper runs both, using the structural result as an independent check.

3. Method

3.1. Forward model and calibration

Melt pool geometry is computed from the Eagar–Tsai solution for a moving Gaussian heat source on a semi-infinite solid, extended with the Gladush–Smurov keyhole depth correction following Honarmandi et al. [13]. Four parameters — effective absorptivity, effective conductivity, heat capacity and the keyhole constant — are fitted to the seven IN718 bare-plate single-track cases of the 2022 Additive Manufacturing Benchmark [12], giving **11.7 % error on melt pool width and 13.0 % on depth**. The keyhole correction is necessary rather than optional: all seven cases have depth-to-half-width aspect ratios between 1.45 and 4.28, and pure conduction gives a 27.8 % depth error on the same cases.

The quadrature in the Eagar–Tsai history integral and the derivative step are both set by convergence tests rather than assumed. Central differences $[y(+h) - y(-h)]/2h$ are evaluated at $h = 0.002$; entries still move by 39.4 % between $h = 0.04$ and $h = 0.02$, fall below 0.3 % at $h = 0.005$, and the residual error follows the second-order scaling the central-difference expansion predicts.

3.2. Fault definitions

Five causes are modelled, each defined as a *physical* quantity acting on a model input so that no arbitrary severity-to-signal constant is introduced. For fault magnitude f: *power* scales laser power, $P(1 + f)$; *speed* scales scan speed, $v(1 + f)$; *gas* is a plume optical depth attenuating effective absorptivity, $A\,e^{-f}$; *geometry* is a preheat rise in units of 100 K, $T_0 + 100f$; and *layer* is a fractional layer-thickness deviation acting through three channels, below. A positive layer fault denotes a *thinner* layer, as produced by recoater short-feed.

The layer fault acts through three channels, two derived from published measurement. Its absorptivity coupling, −0.230, follows from the powder-to-solid absorptivity ratio of approximately two measured calorimetrically at 1070 nm [14] combined with a radiative penetration depth of a few particle diameters; it implies a bare-solid absorptivity of 0.258, consistent with literature values for nickel alloy at 1 µm and with the fitted value. Its powder-melting enthalpy coupling, +0.186, follows from an energy balance in which melting the layer consumes 15.7 % of the absorbed energy, so a thinner layer frees proportionally more for substrate penetration. Its conductivity coupling remains an unmeasured estimate, swept across its plausible range in Section 4.5 with every conclusion invariant to

it. All couplings are applied in exponential form $A\,e^{cf}$, which agrees with $A(1+cf)$ to first order but cannot pass through zero, where the linear form would drive thermal diffusivity negative and the Eagar–Tsai integrand imaginary.

3.3. Sensor models

Three coaxial observables are computed from the temperature field: a band-integrated photodiode (700–1000 nm), melt pool area, and a two-colour pyrometer (750/950 nm) evaluated as the ratio of integrated band powers and inverted for temperature. Spectral radiance follows Planck's law, with temperature capped at the boiling point before any radiance is computed. Four further candidate observables are defined for Section 4.3: the camera's width and length axes reported separately, a short-wave infrared photodiode, and a wide-band pyrometer.

3.4. Diagnosability measures

The relative-sensitivity Jacobian is

$$J_{sk} \;=\; \frac{1}{y_s}\,\frac{\partial y_s}{\partial f_k}, \tag{1}$$

so entry (s,k) is the fractional change in sensor s per unit of fault k, making rows with different physical units comparable. Each row is divided by that sensor's relative noise σ_s to give the whitened matrix J^{w}.

Three quantities are reported throughout. The *angle* between whitened columns, $\theta_{kl} = \arccos\left(|\langle J^{\mathrm{w}}_{:k}, J^{\mathrm{w}}_{:l}\rangle| \,/\, \|J^{\mathrm{w}}_{:k}\|\|J^{\mathrm{w}}_{:l}\|\right)$, measures pairwise separability: 0° means the two faults move the readings along one line and are indistinguishable by any estimator, 90° that they are orthogonal. A practical floor of 10° is adopted as a convention and its influence reported rather than assumed. The *rank* of J^{w} counts how many fault directions are observable at all, and the *condition number* — the ratio of largest to smallest non-zero singular value — how thin the weakest is; rank without it is misleading, as Section 4.3 shows. Structural analysis is run in parallel on an eleven-equation representation of the same model using MSO sets, as an independent check.

3.5. Build context and the enlarged observation

A build is not a single instant. It comprises thousands of segments whose toolpath, layer index, gas-flow direction and local geometry are known to the machine, and the five faults do not vary identically across them: laser degradation drifts with time, galvanometer error concentrates at direction reversals, a recoater short-feed recurs with layer index, gas degradation follows the flow direction, and heat accumulation follows support depth. Treat the build, not the instant, as the observation: let $G \in \mathbb{R}^{N\times 5}$ hold five known context regressors over N segments — a constant, position along the gas-flow axis, layer parity, turn density and support depth — and $M \in \mathbb{R}^{N\times 5}$ the spatial modulation profile of each fault. Regressing each sensor's relative residual onto G yields five coefficients per sensor in place of one reading, so three sensors give $3 \times 5 = 15$ observables, with Jacobian

$$J^{\mathrm{big}}_{(s,j),k} = J_{sk}\,B_{jk}, \qquad B = (G^{\top}G)^{-1}G^{\top}M. \tag{2}$$

The noise on the j-th coefficient from sensor s is not σ_s but $\sigma_{(s,j)} = \sigma_s\sqrt{[(G^{\top}G)^{-1}]_{jj}}$, which here lies between 0.109 and 0.136, so fitting $N = 84$ segments to five regressors buys roughly a ninefold noise reduction. Results use $N = 84$; the dependence on N is given in Section 4.5.

The profiles in M are physical claims rather than measurements, and one has direct experimental support. The AM Bench pad scans [12], under a 4.3 m s^{-1} cross-flow, show melt pool depth reduced by 27.2 % on average (range 19.4–31.5 % over four cross-sections) on tracks scanned parallel to the gas flow relative to anti-parallel tracks, against expanded uncertainties of 6–11 µm on depths near 175 µm, attributed to vapour-plume interaction with the laser. The measurement carries its own control: on pads scanned perpendicular to the flow the odd–even difference is 2.4–4.7 %, within or near uncertainty. A deterministic gas-related modulation of melt pool geometry, tied to a known toolpath quantity, is therefore established by measurement rather than asserted.

Two qualifications follow from the same source. The measured modulation is organised by scan direction relative to the gas flow rather than by position along it, and it alternates at track scale where a recoater signature alternates at layer scale — so the two have distinct periodicities, which Section 4.5 shows is the condition the method actually requires.

3.6. Position on sensor noise

Only one in-process noise figure proved citable: a melt pool width uncertainty of 2.3 % relative, from the uncertainty budget of Weaver et al. [12]. No comparable in-process characterisation was found for a coaxial photodiode or two-colour pyrometer; published thermal calibration work for such sensors reports instrument calibration, not in-process noise. Sensor noise is therefore treated as a *ratio* and swept over four assumption sets, with every conclusion required to hold under all of them. This suffices for isolability, which depends only on relative noise — angles, rank and condition number are invariant to a common scaling of all sensor noise — so absolute detection thresholds are **not** claimed.

4. Results

4.1. Validation

Two checks are made against data not used in the fit. Over 27 melted conditions the model reproduces the benchmark correlation structure: depth correlates with volumetric energy density ($r = +0.727$) while width does not ($r = +0.032$) [12]. Against the constant-energy-density experiment of Guo et al. [11] the model achieves a **partial pass**: it reproduces the existence, direction and regime transition of the variation, but spans 2.37× in depth where the experiment reports orders of magnitude. The mechanism Guo et al. identify — absorption varying with vapour depression geometry — is absent from a conduction solution with an analytic keyhole patch, and this is reported as a model limitation. The same sweep yields a result used in Section 4.7: at constant energy density melt pool *length* varies by 2.53× while width and depth vary by only 1.11×.

4.2. *Laser power and shielding gas are exactly confounded*

Both faults act *only* on the product (effective absorptivity × power), and the Eagar–Tsai temperature rise is linear in that product. Choosing $f_{\text{gas}} = -\ln(1 + f_{\text{power}})$ therefore produces an identical melt pool: the same temperature at every point, hence identical readings from any instrument that could be built. The ratio of the two Jacobian columns is −0.99999, −1.00000, −0.99993 across the three sensors, a spread of 7×10^{-5} attributable to finite-difference error on an exact identity. The pairwise angle is **0.00**°. This is an algebraic degeneracy, not a numerical coincidence: sweeping every coupling constant across its plausible range leaves the angle at 0.00°, and the independent structural analysis identifies the same pair from equation form alone.

4.3. *The limit is the melt pool, not the sensor count*

The whitened 3×5 Jacobian has singular values 3.451, 0.155 and 0.041 — rank 3, condition number 85 — so two fault directions are unobservable, and they are computable. The dominant one is (power+0.670, gas+0.731, layer−0.112, speed−0.056); applied as a real fault of 5 % vector norm it moves the sensors by −0.083 %, −0.067 % and −0.002 %, against +7.03 %, +6.02 % and +0.049 % for a plain 5 % power fault — a factor of 85, the condition number.

Adding all four candidate observables gives seven rows and raises the rank only from 3 to **4**. The fourth direction is of no practical use: the singular values become (4.58, 0.475, 0.199, 2.0×10^{-5}, 3.9×10^{-6}), a condition number of 2.3×10^{5}, and the worst pairwise angle remains 0.00°. Seven instruments leave power and gas exactly as confounded as one.

The reason is structural. Everything the model knows funnels into four physical channels — absorbed power, scan speed, effective conductivity and preheat — with three of the five faults crowding onto the first. Any instrument reading only the melt pool is a function of those four channels, so the fault-to-sensor map cannot exceed rank 4 however many are fitted: a new wavelength reweights the same Planck curve over the same pool, adding a row, not a direction.

4.4. *Build context dissolves the limit*

With the enlarged observation of Section 3.5, rank rises from 3 to **5**, the worst pair from 0.00° to **25.83**°, power versus gas to 29.34°, and the condition number falls from 85 to 30. All ten pairs clear the 10° floor (Fig. 1).

Three controls establish that this arises from the claimed mechanism, not the arithmetic. First and most important, giving all five faults one identical profile returns the instantaneous result *exactly* — rank 3, condition number 85, worst pair 0.00° — despite the enlarged matrix still having 15 rows, because proportional columns of B make the enlarged Jacobian inherit the instantaneous column directions. **The gain is not dimensionality, it is the profiles differing.** Second, replacing every derivative by a secant through the full nonlinear chain gives rank 5 at every magnitude from 0.2 % to 20 % in both signs, worst pair 22.19°–36.90°, condition number 30–43; the derivative limit is *not* strictly conservative, a −2 % fault giving 22.19°, still 2.2× the floor. Third, recomputing that study with the full model at every segment and no interpolation reproduces the angles to 0.10°.

4.5. *Robustness*

Because the profiles are asserted, separability is swept against profile contrast. Eight of ten pairs clear the floor at or below 25 % of the asserted strength and the two hardest, power/speed and power/gas, at **40 %**: the conclusion survives the real profiles being 2.5× weaker than claimed.

Four further sweeps are reported. All 120 permutations of which fault carries which profile give rank 5, worst angle 17.13°–30.58°, so the *assignment* of profiles does not matter. The unmeasured conductivity coupling leaves the worst angle at 25.83° throughout its range, and four noise-ratio assumption sets give 24.97°–25.83°. Segment count gives 11.74° at $N = 12$, 23.91° at $N = 30$, 25.83° at $N = 84$ and 23.03° at $N = 2400$: of order 30 segments are required, and more adds nothing.

What the method does require is that the profiles be *distinct*. Forcing pairs equal identifies two conditions on which the result depends: if the recoater and gas profiles coincide the worst angle collapses to 3.84°, and if the speed and gas profiles coincide, to 8.12°. Both are below the floor and both would defeat the method. Every other collapse is survivable, and anything sharing a profile with the geometry fault leaves the answer unchanged. The first condition is mild — the recoater advances once per layer along a fixed axis while the gas flows along a fixed chamber axis, and the benchmark measurement of Section 3.5 confirms the two signatures have different periodicities. The second constrains the scan strategy rather than the machine, and is the weaker of the two.

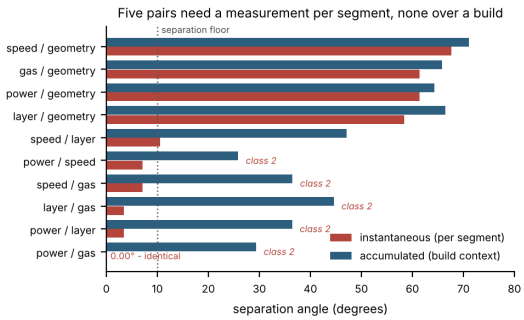


Figure 1: Diagnosability matrix for the baseline coaxial suite in both regimes. Per-segment decisions leave five of ten pairs requiring an added measurement; accumulated diagnosis over a build resolves all ten from existing signals. Class 3, structurally irreducible, is empty in both regimes.

4.6. *Diagnosability matrix and configuration classes*

Classifying on the enlarged observation alone places all ten pairs in class 1 for every configuration, including a single photodiode — true but uninformative. The matrix is therefore reported in two regimes: *instantaneous*, per-segment decisions for closed-loop control, and *accumulated*, diagnosis over a build for qualification and drift monitoring. The baseline suite

gives five of ten pairs in class 1 instantaneously and ten of ten accumulated, with **class 3 empty in both regimes** (Fig. 1): nothing is structurally irreducible for this suite.

No vendor is named in Table 1: mapping a product to a set of observables rests on secondary literature rather than datasheets, since detection bands are routinely withheld and published descriptions of the same product disagree, so attaching a product name to a computed limit would assert more than the evidence carries. Stripping the names also reveals what a vendor-by-vendor table hides — across five commercial systems surveyed there are only three distinct configurations for this analysis.

Table 1: Diagnosability by configuration class. Class 1 counts pairs resolvable from existing signals; the last column is the instantaneous power/gas angle. E is a configuration type, not a surveyed product.

	Configuration	Rank	Class 1	power/gas
A	single coaxial photodiode	1	0/10	0.00°
B	photodiode + pool imaging	2	4/10	0.00°
C	two photodiodes, separate bands	2	0/10	0.00°
D	research reference (B + pyrometry)	3	5/10	0.00°
E	C + upstream power monitoring	3	4/10	**84.02**°

No configuration built solely from melt-pool observables separates laser power from gas degradation, whatever its sensor count: A through D all read 0.00°, for the reason given in Section 4.3. Configuration E is the actionable row — a direct measurement of delivered laser power sampled *upstream* of the process zone separates the pair at 84.02°, because such a sensor sees the laser output without the plume attenuation. One surveyed system may belong at E rather than C, depending on whether its laser-energy photodiode samples upstream; this could not be established from public documentation, so it is counted conservatively as C. Note also that C performs worse than B despite both having two observables: two bands of the same emitted light are near-redundant, whereas a geometric measurement adds a direction.

4.7. What an added measurement would buy

The sensors most likely to help — plume sensing for the gas fault, powder-bed or recoater sensing for the layer fault [8, 9] — cannot be simulated by a model with neither a plume nor a powder layer. What such a sensor *measures* can be: a unit row on fault k with a given relative noise, the idealised best case for any instrument aimed at that cause.

A direct gas measurement clears power/gas, speed/gas and layer/gas; a direct geometry measurement clears nothing and leaves the rank at 3, geometry being already separated at 58–68°. **Precision is almost irrelevant**: sweeping the added sensor's relative noise from 50 % to 0.2 % leaves the worst pair at 3.46° and the cleared count at 8 of 10 throughout, because adding a row in a new direction changes the rank while making an existing row quieter does not. A crude measurement of the right cause beats a precise measurement of the wrong one.

The comparison that matters follows. **Build context outperforms two idealised, perfect direct measurements of the causes themselves**: gas and layer sensors together give rank 5 but a worst pair of 7.12°, condition number 546 and nine of ten pairs cleared, against 25.83°, 30 and ten of ten for build context at no hardware cost. Neither added sensor says anything about power versus speed, and that pair is exactly what spatial context resolves.

Among melt-pool observables the most valuable addition is free. Reporting the camera's two axes separately rather than as an area is worth **+13.40**° on the binding pair, raising power/speed from 25.83° to 39.23° and the overall worst pair to 29.34°, whereas a short-wave infrared photodiode (−0.08°) and a wide-band pyrometer (−0.13°) are worth nothing. The mechanism is the validation result above: length carries 2.5× the variation of width and depth at constant energy density, and an area multiplies the least informative dimension by the most informative one.

5. Discussion

5.1. Relation to reported monitoring performance

The premise that predicted limits could be compared against reported accuracies turns out to be mostly unrealisable, and the reason is itself the result. Of 18 works extracted, 12 classify outcomes, about which this analysis makes no prediction: a melt pool deeper than the layer will fuse whatever made it deep. No published work attempts to separate power from gas, or layer from power — the pairs identified here as confounded. That is absence of evidence rather than evidence of absence, but no published result contradicts the predicted limits.

Where comparison is possible it is consistent. Two independent groups report the lumped group this analysis identifies as what survives confounding: RAISE-LPBF ranks on linear power density as a target in its own right [5], and parameter estimation from melt pool imagery predicts energy density class [10]. One quantitative comparison remains available and should be made before any strong claim of model fidelity. Reijonen reports the photodiode response to scan speed as a power law $y = ax^{-b}$ [6], whose relative sensitivity is exactly $-b$ and so is directly comparable to the Jacobian entry here, −0.483; the same work reports the power response as approximately linear where the entry here, +1.518, is superlinear. Whether these agree is a decisive test of the forward model.

5.2. A design rule

The failed and successful candidates differ by a criterion that can be stated generally: *a sensor added to break a confound must observe upstream of the shared channel; a sensor reading the melt pool adds a row, not a direction.* This explains why two photodiodes in different bands (C) buy nothing while an upstream power monitor (E) buys 84°, and why plume and powder-bed sensing are the right classes for the gas and layer faults. Before buying an instrument, extract more independent directions from those already fitted.

5.3. Limitations

The thermal model is a conduction solution with an analytic keyhole patch: it partial-passes the Guo validation, absolute magnitudes are not claimed, and the fitted parameters sit at their bounds. The absorptivity, conductivity and heat capacity correlations of 0.93–0.96 reported during Bayesian calibration of the same model family [13] are the same pathology this paper studies, appearing in its own calibration. One coupling remains an unmeasured estimate, and no benchmark dataset with deliberately induced machine faults exists against which to calibrate any of them, so each is defended by sweeping rather than by argument.

Three modelling choices make the instantaneous case *more* confounded than reality, and so inflate the gap by which the contribution is measured. Defining the gas fault as pure optical depth omits a convective channel a real gas fault would have. Treating plume attenuation as independent of power and speed is what makes the power/gas degeneracy exact, whereas the photodiode response to gas flow is reported to depend interactively on the other parameters [6], and reduced gas flow causes loss of penetration through plume obstruction [7]. Modelling upstream power monitoring as a melt-pool-derived signal is likewise pessimistic. **The reported** 0.00° **should be read as an upper bound on the instantaneous confounding.** The rank limit is unaffected, deriving from channel counting rather than from the form of this coupling.

The modulation profiles are the largest remaining exposure. One is supported by benchmark measurement, but by a measurement organised differently from the regressor used here; the others are asserted. The contrast and permutation sweeps are the defence, and a partial one: they establish how much a wrong profile costs, not that the assumed profiles are right. Finally, the five fault magnitudes are physical quantities but not on a common severity scale, so comparisons *across pairs at fixed magnitude* are not meaningful. All results are diagnosability statements — what a suite can distinguish in principle, at a given relative noise, under a given forward model — not that any algorithm achieves it.

6. Conclusions

Laser power deviation and shielding gas degradation produce identical melt pools in this model, confounded at 0.00° by algebra rather than by noise, so no instrument reading the melt pool separates them from a single instant. The binding limit is the melt pool, which offers four physical channels, not the instrumentation: seven observables raise the attainable rank only from 3 to 4, at a condition number of 2.3×10^5, leaving the worst pair at 0.00°.

The structure already present in a build dissolves the limit at no hardware cost — rank $3 \rightarrow 5$, worst pair $0.00^\circ \rightarrow 25.83^\circ$ — surviving at 40 % of the asserted profile strength and at fault magnitudes from 0.2 % to 20 %, with a shared-profile control reproducing the instantaneous result exactly and so confirming the mechanism. That free method outperforms two idealised direct measurements of the causes themselves, because neither addresses power versus speed, and it requires only that the recoater and gas signatures, and the speed and gas signatures, vary differently across a build.

No configuration built from melt-pool observables alone separates power from gas instantaneously; an upstream measurement of delivered laser power does, at 84°, and that is the recommendation for real-time control. Among melt-pool observables the most valuable addition is free — reporting the camera's two axes separately is worth $+13.40^\circ$ on the binding pair, while two further instruments are worth nothing. Nothing is structurally irreducible for this suite: post-build diagnosis needs no new hardware, real-time per-segment isolation does, and crude sensing of the right cause suffices. The method needs only a forward model and a sensor description, and answers at specification time a question otherwise settled after the hardware is bought.

References

[1] M. Krysander, E. Frisk, Sensor placement for fault diagnosis, IEEE Trans. Syst. Man Cybern. A 38 (6) (2008) 1398–1410.

[2] E. Frisk, M. Krysander, D. Jung, A toolbox for analysis and design of model based diagnosis systems for large scale models, IFAC-PapersOnLine (2017). doi:10.1016/j.ifacol.2017.08.504.

[3] L. Scime, J. Beuth, Using machine learning to identify in-situ melt pool signatures indicative of flaw formation in a laser powder bed fusion additive manufacturing process, Addit. Manuf. 25 (2019) 151–165.

[4] A. Raj, Y. Huang, R. Stegman, et al., Modeling spatial variations in co-axial melt pool monitoring signals in laser powder bed fusion, J. Manuf. Process. 89 (2023) 24–38.

[5] C. Blanc, A. Ahar, K. De Grave, Reference dataset and benchmark for reconstructing laser parameters from on-axis video in powder bed fusion of bulk stainless steel, arXiv preprint arXiv:2409.12475 (2024).

[6] J. Reijonen, Effect of laser parameters and shielding gas flow on co-axial photodiode-based melt pool monitoring signals in laser powder bed fusion, Addit. Manuf. Lett. 11 (2024) 100244.

[7] J. Reijonen, A. Revuelta, T. Riipinen, et al., On the effect of shielding gas flow on porosity and melt pool geometry in laser powder bed fusion additive manufacturing, Addit. Manuf. 32 (2020) 101030.

[8] T. Liu, C. S. Lough, H. Sehhat, et al., In-situ infrared thermographic inspection for local powder layer thickness measurement in laser powder bed fusion, Addit. Manuf. 55 (2022) 102873.

[9] M. Bugatti, B. M. Colosimo, The intelligent recoater: A new solution for in-situ monitoring of geometric and surface defects in powder bed fusion, Addit. Manuf. Lett. 3 (2022) 100048.

[10] Estimation of laser powder bed fusion process parameters from melt pool images using k-nearest neighbours, Int. J. Adv. Manuf. Technol. (2025). doi:10.1007/s00170-025-15591-y.

[11] Q. Guo, C. Zhao, M. Qu, et al., In-situ characterization and quantification of melt pool variation under constant input energy density in laser powder bed fusion additive manufacturing process, Addit. Manuf. 28 (2019) 600–609.

[12] J. S. Weaver, D. Deisenroth, S. Mekhontsev, et al., Cross-sectional melt pool geometry of laser scanned tracks and pads on nickel alloy 718 for the 2022 additive manufacturing benchmark challenges, Integr. Mater. Manuf. Innov. 13 (2024) 363–379.

[13] P. Honarmandi, R. Seede, L. Xue, et al., A rigorous test and improvement of the Eagar-Tsai model for melt pool characteristics in laser powder bed fusion additive manufacturing, Addit. Manuf. 47 (2021) 102300.

[14] J. Trapp, A. M. Rubenchik, G. Guss, M. J. Matthews, In situ absorptivity measurements of metallic powders during laser powder-bed fusion additive manufacturing, Appl. Mater. Today (2017). doi:10.1016/j.apmt.2017.08.006.